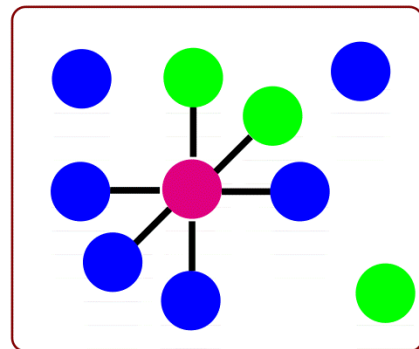
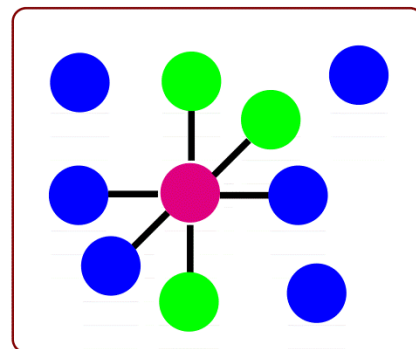
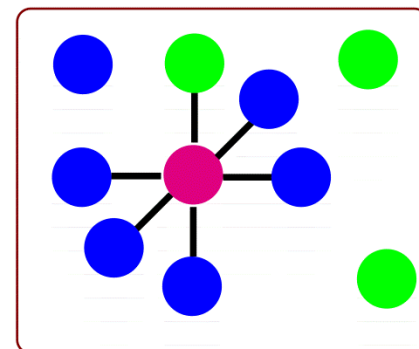
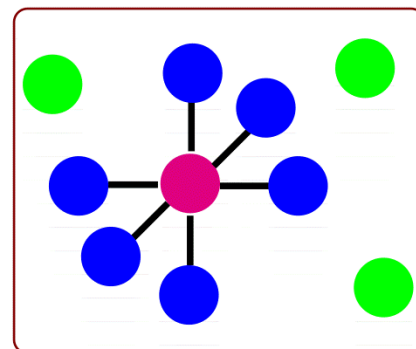
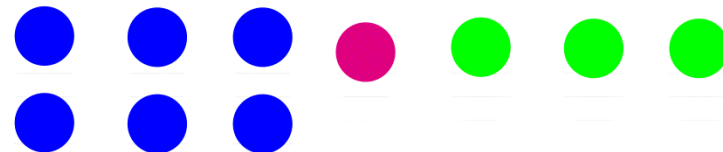
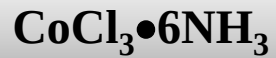


Crystal Field Theory and Ligand Field Theory

Werner's Coordination Theory (1893)

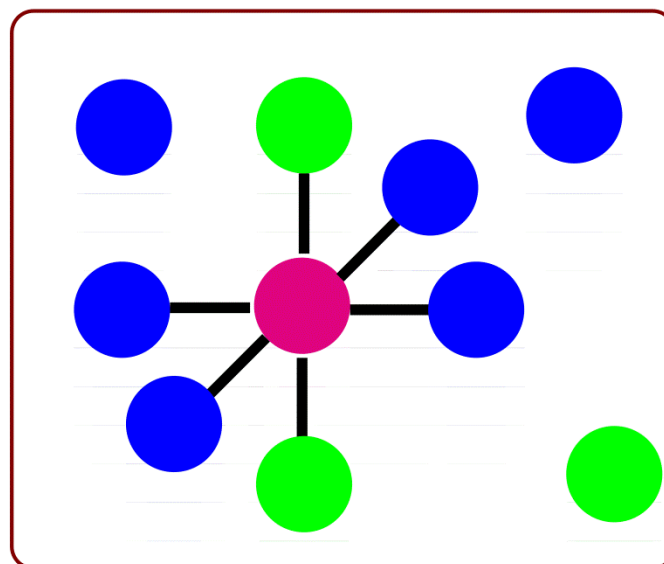
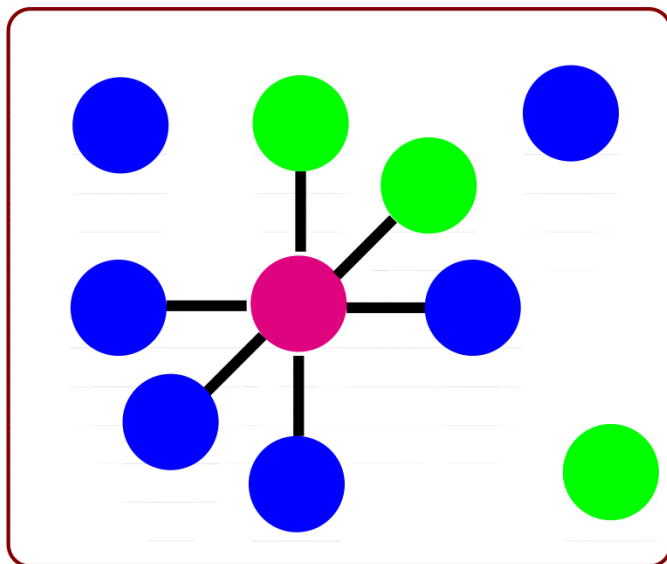


Double salt??

Primary vs. Secondary Coordination

TABLE 24.1 Properties of Some Ammonia Complexes of Cobalt(III)

Original Formulation	Color	Ions per Formula Unit	"Free" Cl^- Ions per Formula Unit	Modern Formulation
$\text{CoCl}_3 \cdot 6 \text{NH}_3$	Orange	4	3	$[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$
$\text{CoCl}_3 \cdot 5 \text{NH}_3$	Purple	3	2	$[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$
$\text{CoCl}_3 \cdot 4 \text{NH}_3$	Green	2	1	<i>trans</i> - $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}$
$\text{CoCl}_3 \cdot 4 \text{NH}_3$	Violet	2	1	<i>cis</i> - $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}$

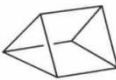


Valence Bond Theory (1928-1930)

Table 20.1 Hybridization schemes for the σ -bonding frameworks of different geometrical configurations of ligand donor atoms.

Coordination number	Arrangement of donor atoms	Orbitals hybridized	Hybrid orbital description	Example
2	Linear	s, p_z	sp	$[\text{Ag}(\text{NH}_3)_2]^+$
3	Trigonal planar	s, p_x, p_y	sp^2	$[\text{HgI}_3]^-$
4	Tetrahedral	s, p_x, p_y, p_z	sp^3	$[\text{FeBr}_4]^{2-}$
4	Square planar	$s, p_x, p_y, d_{x^2-y^2}$	$sp^2 d$	$[\text{Ni}(\text{CN})_4]^{2-}$
5	Trigonal bipyramidal	$s, p_x, p_y, p_z, d_{z^2}$	$sp^3 d$	$[\text{CuCl}_5]^{3-}$
5	Square-based pyramidal	$s, p_x, p_y, p_z, d_{x^2-y^2}$	$sp^3 d$	$[\text{Ni}(\text{CN})_5]^{3-}$
6	Octahedral	$s, p_x, p_y, p_z, d_{z^2}, d_{x^2-y^2}$	$sp^3 d^2$	$[\text{Co}(\text{NH}_3)_6]^{3+}$
6	Trigonal prismatic	$s, d_{xy}, d_{yz}, d_{xz}, d_{z^2}, d_{x^2-y^2}$ or $s, p_x, p_y, p_z, d_{xz}, d_{yz}$	sd^5 or $sp^3 d^2$	$[\text{ZrMe}_6]^{2-}$
7	Pentagonal bipyramidal	$s, p_x, p_y, p_z, d_{xy}, d_{x^2-y^2}, d_{z^2}$	$sp^3 d^3$	$[\text{V}(\text{CN})_7]^{4-}$
7	Monocapped trigonal prismatic	$s, p_x, p_y, p_z, d_{xy}, d_{xz}, d_{z^2}$	$sp^3 d^3$	$[\text{NbF}_7]^{2-}$
8	Cubic	$s, p_x, p_y, p_z, d_{xy}, d_{xz}, d_{yz}, f_{xyz}$	$sp^3 d^3 f$	$[\text{PaF}_8]^{3-}$
8	Dodecahedral	$s, p_x, p_y, p_z, d_{z^2}, d_{xy}, d_{xz}, d_{yz}$	$sp^3 d^4$	$[\text{Mo}(\text{CN})_8]^{4-}$
8	Square antiprismatic	$s, p_x, p_y, p_z, d_{xy}, d_{xz}, d_{yz}, d_{x^2-y^2}$	$sp^3 d^4$	$[\text{TaF}_8]^{3-}$
9	Tricapped trigonal prismatic	$s, p_x, p_y, p_z, d_{xy}, d_{xz}, d_{yz}, d_{z^2}, d_{x^2-y^2}$	$sp^3 d^5$	$[\text{ReH}_9]^{2-}$

Coordination

Number	Geometry	Polyhedron	Comments
1		—	Unimportant
2 Linear		—	Uncommon: found mainly with d^{10} metal ions
3 Trigonal plane			Rare; can be induced by use of sterically bulky ligands
4 Square plane			Common for d^8 metal ions otherwise unusual;
4 Tetrahedron			Fairly common, especially for d^{10} and some d^5 ions
5 Trigonal bipyramid			Rare Examples are often similar in structure and energy so may easily interconvert
5 Square pyramid			
6 Octahedron			Very common; usually the most favoured energetically and gives the lowest ligand–ligand repulsions
(Octahedron = trigonal antiprism)			An alternative view of an octahedron down a three-fold rotation axis
6 Trigonal prismatic			Rare, and requires some extra steric or electronic benefit to be favoured over octahedral

Coordination

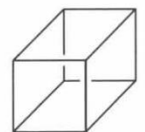
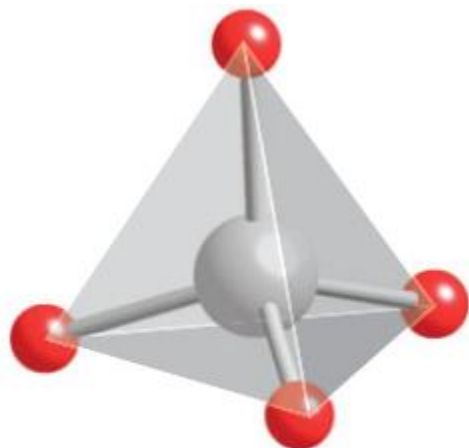
Number	Geometry	Polyhedron	Comments
7 Pentagonal bipyramid			Uncommon
7 Monocapped octahedron			Uncommon
8 Dodecahedron			Most sterically efficient geometric arrangement for eight equivalent ligands
8 Square antiprism			Uncommon
8 Cube			Rare; found only with the largest metal ions
8 Hexagonal bipyramid			Quite common for eight-coordinate complexes of metals with <i>trans</i> -dioxo ligands

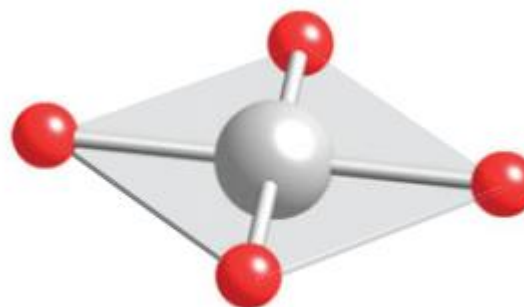
Table 16.5 A Summary of Types of Complexes Formed by First-Row Metal Ions.

Number of <i>d</i> Electrons	Most Common Ion	Usual Geometry ^b	Hybrid Orbital	Example
0	Sc ³⁺	Octahedral	<i>d</i> ² <i>sp</i> ³	Sc(H ₂ O) ₆ ³⁺
1	Ti ³⁺	Octahedral	<i>d</i> ² <i>sp</i> ³	Ti(H ₂ O) ₆ ³⁺
2	V ³⁺	Octahedral	<i>d</i> ² <i>sp</i> ³	VF ₆ ³⁻
3	Cr ³⁺	Octahedral	<i>d</i> ² <i>sp</i> ³	Cr(NH ₃) ₆ ³⁺
4 ^a	Mn ³⁺	Octahedral (h.s.)	<i>sp</i> ³ <i>d</i> ²	Mn(H ₂ O) ₆ ³⁺
4 ^a	Mn ³⁺	Octahedral (l.s.)	<i>d</i> ² <i>sp</i> ³	Mn(CN) ₆ ³⁻
5	Fe ³⁺	Octahedral (h.s.)	<i>sp</i> ³ <i>d</i> ²	Fe(H ₂ O) ₆ ³⁺
5	Fe ³⁺	Octahedral (l.s.)	<i>d</i> ² <i>sp</i> ³	Fe(CN) ₆ ³⁻
5	Fe ³⁺	Tetrahedral	<i>sp</i> ³	FeCl ₄ ⁻
6	Co ³⁺	Octahedral (h.s.)	<i>sp</i> ³ <i>d</i> ²	CoF ₆ ³⁻
6	Co ³⁺	Octahedral (l.s.)	<i>d</i> ² <i>sp</i> ³	Co(H ₂ O) ₆ ³⁺
7	Co ²⁺	Octahedral (h.s.)	<i>sp</i> ³ <i>d</i> ²	Co(H ₂ O) ₆ ²⁺
7	Co ²⁺	Trigonal bipyramid	<i>dsp</i> ³	Co(CN) ₅ ³⁻
7	Co ²⁺	Tetrahedral	<i>sp</i> ³	CoCl ₄ ²⁻
7	Co ²⁺	Square planar	<i>dsp</i> ²	Co(CN) ₄ ²⁻
8	Ni ²⁺	Tetrahedral	<i>sp</i> ³	Ni(NH ₃) ₄ ²⁺
8	Ni ²⁺	Octahedral	<i>sp</i> ³ <i>d</i> ²	Ni(NH ₃) ₆ ²⁺
8	Ni ²⁺	Trigonal bipyramid	<i>dsp</i> ³	Ni(CN) ₅ ³⁻
8	Ni ²⁺	Square base pyramid	<i>sp</i> ² <i>d</i> ²	Ni(CN) ₅ ³⁻
8	Ni ²⁺	Square planar	<i>dsp</i> ²	Ni(CN) ₄ ²⁻
9 ^a	Cu ²⁺	Octahedral	<i>sp</i> ³ <i>d</i> ²	CuCl ₆ ⁴⁻
9 ^a	Cu ²⁺	Tetrahedral	<i>sp</i> ³	Cu(NH ₃) ₄ ²⁺
10	Zn ²⁺	Octahedral	<i>sp</i> ³ <i>d</i> ²	Zn(H ₂ O) ₆ ²⁺
10	Zn ²⁺	Tetrahedral	<i>sp</i> ³	Zn(NH ₃) ₄ ²⁺
10	Ag ⁺	Linear	<i>sp</i>	Ag(NH ₃) ₂ ⁺

^aIrregular structure due to Jahn-Teller distortion.^bh.s. and l.s. indicate high and low spin, respectively.

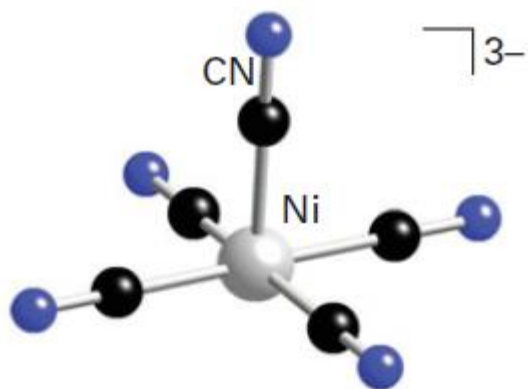


14 Tetrahedral complex (T_d)



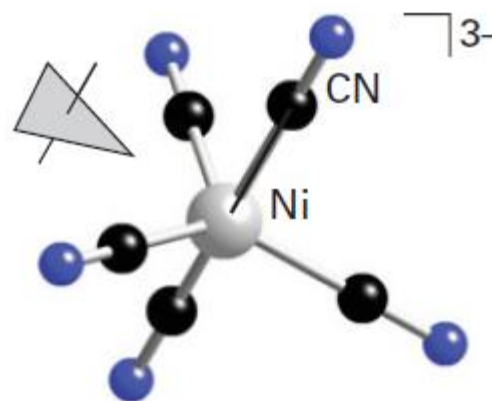
15 Square-planar complex (D_{4h})

$\tau_4?$

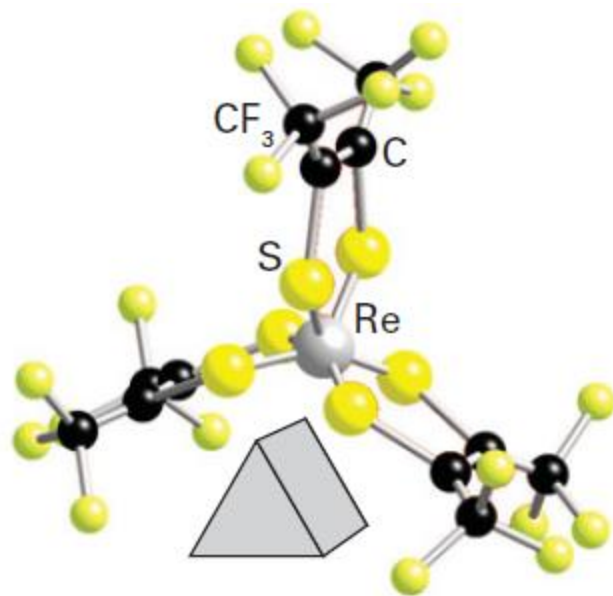


16 $[\text{Ni}(\text{CN})_5]^{3-}$, square pyramidal (C_{4v})

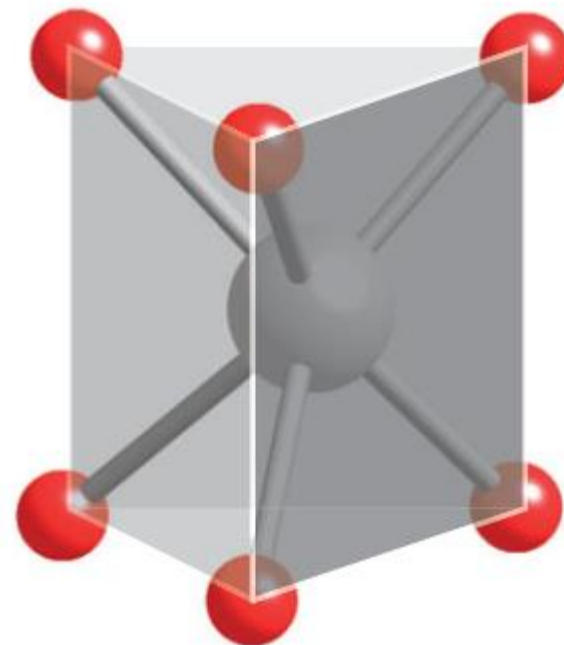
$\tau_5?$



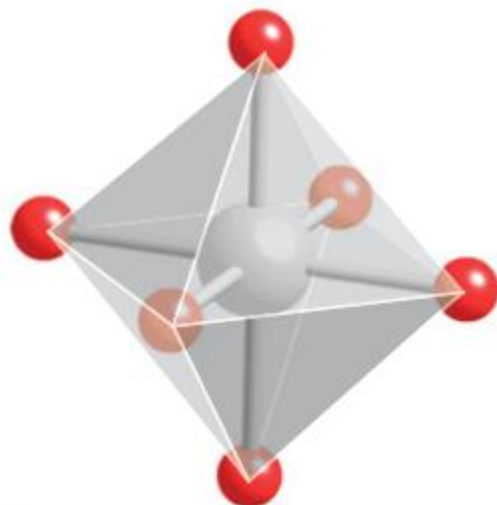
51 $[\text{Ni}(\text{CN})_5]^{3-}$, trigonal bipyramidal (D_{3h})



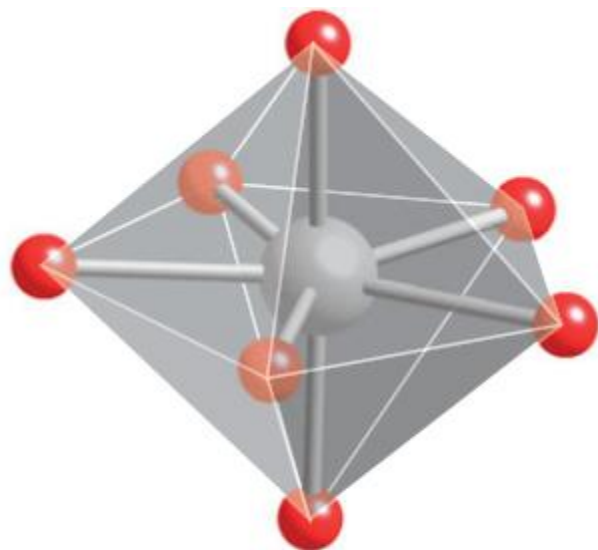
21 $[\text{Re}(\text{S}(\text{CF}_3)\text{C}=\text{C}(\text{CF}_3)\text{S})_3]$



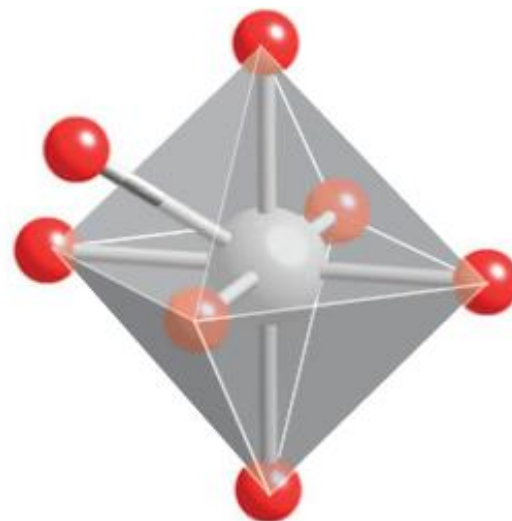
20 Trigonal prism (D_{3h})



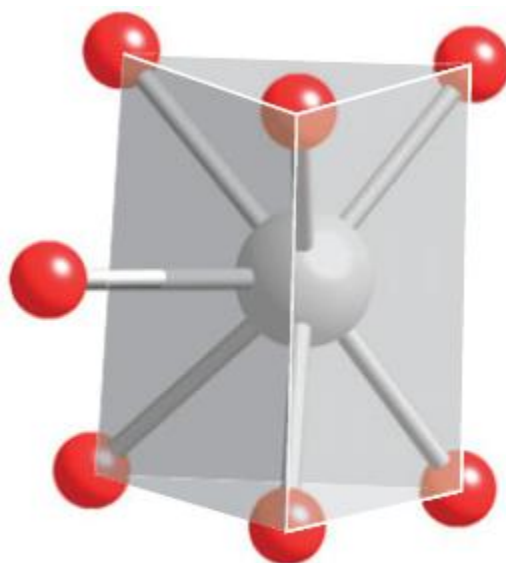
19 Octahedral complex (O_h)



22 Pentagonal bipyramid (D_{5h})



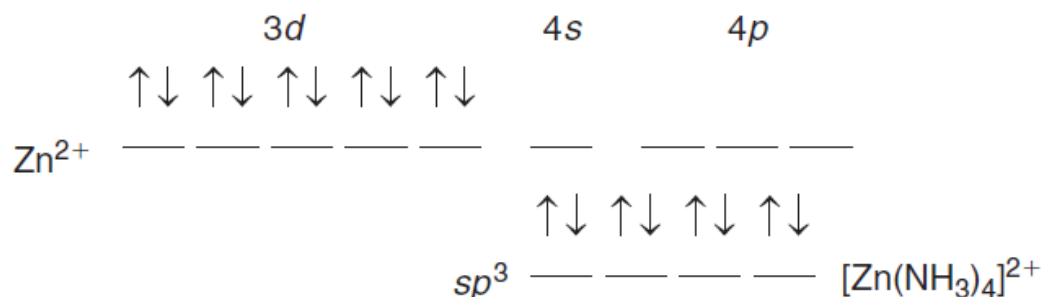
23 Capped octahedron



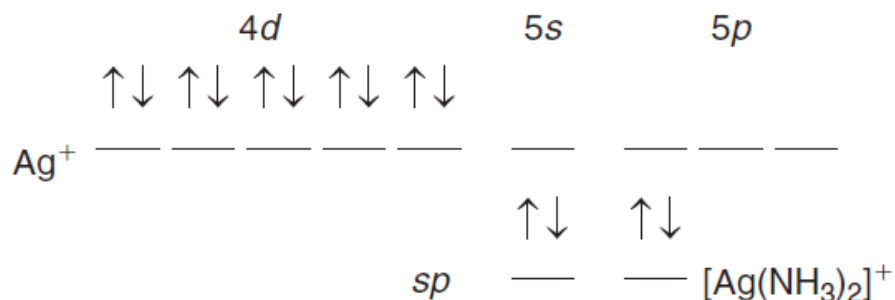
24 Capped trigonal prism

Hybridization

When Zn^{2+} forms a complex such as $[\text{Zn}(\text{NH}_3)_4]^{2+}$, it is easy to rationalize the bonding in terms of hybrid orbitals. Zn^{2+} has a d^{10} configuration, but the 4s and 4p orbitals are empty. Therefore, a set of four empty sp^3 hybrids can result. As shown next, four orbitals can accept four pairs of electrons that are donated by the ligands.

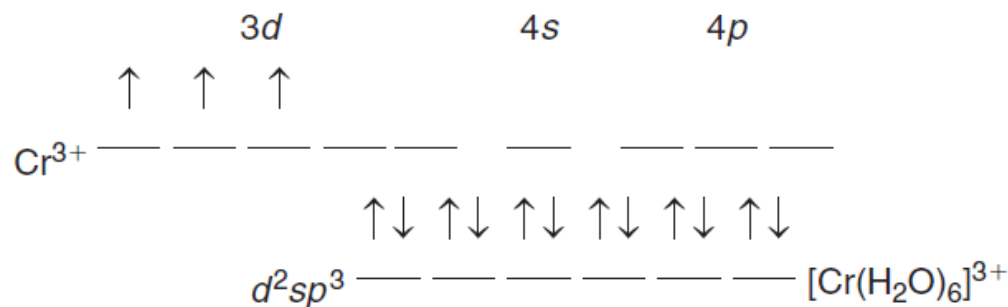


Likewise, it is easy to rationalize how Ag^+ (a d^{10} ion) can form a linear complex with two ligands like NH_3 .

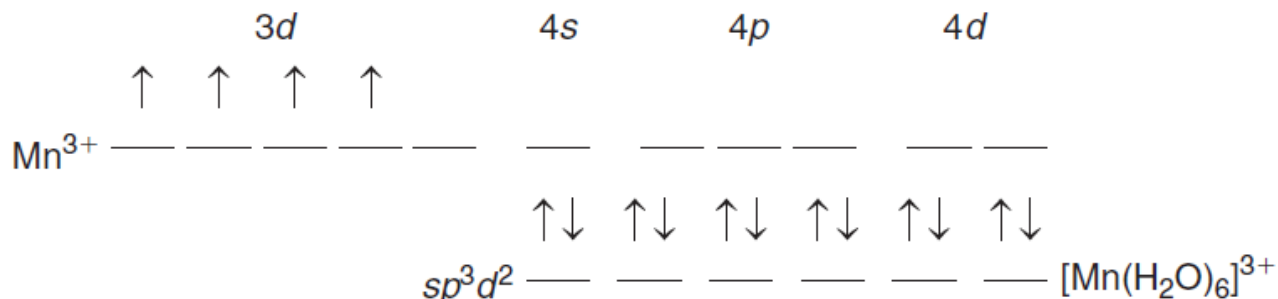


Hybridization

For metal ions having configurations d^0 , d^1 , d^2 , or d^3 , there will always be two of the d orbitals empty to form a set of d^2sp^3 hybrids. Therefore, we expect complexes of these metal ions to be octahedral in which the hybrid orbital type is d^2sp^3 . If we consider Cr^{3+} as an example, the formation of a complex can be shown as follows:

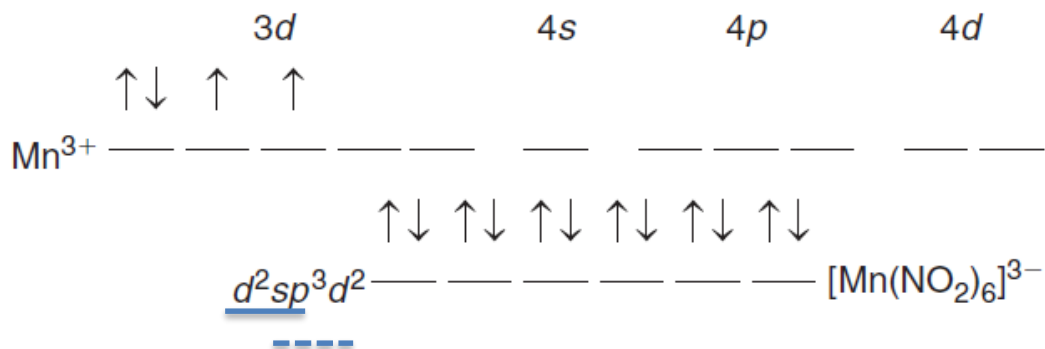


When the number of electrons in the d orbitals is four, as in the case of Mn^{3+} , there exists more than one possible type of hybrid orbital. For example, if the electrons remain unpaired in the d orbitals, there is only one orbital in the set that is empty. As a result, if an octahedral complex is formed, making use of two d orbitals requires that the $4d$ orbitals be used so that sp^3d^2 hybrid orbitals would be used by the metal. This case can be shown as follows:

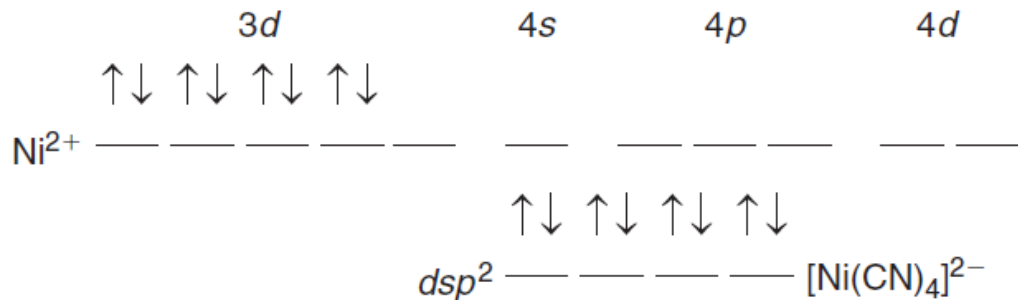


Hybridization

This bonding arrangement results in four unpaired electrons in the $3d$ orbitals, but with some ligands, the situation is different. In those cases, two of the $3d$ orbitals are made available to form a set of hybrid orbitals by means of electron pairing. When this occurs, the bonding arrangement can be shown as follows:

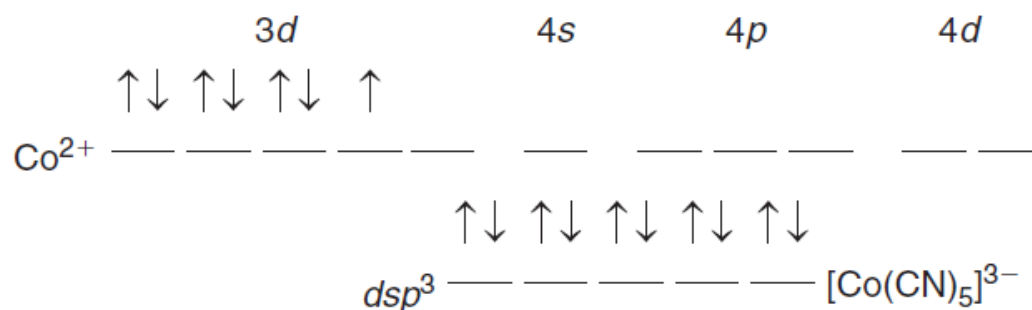


If a complex having a coordination number of 4 is produced, only two of the $4p$ orbitals are used and the hybrid orbital type is dsp^2 , which is characteristic of a square planar complex. For Ni^{2+} , this is illustrated by the following scheme:

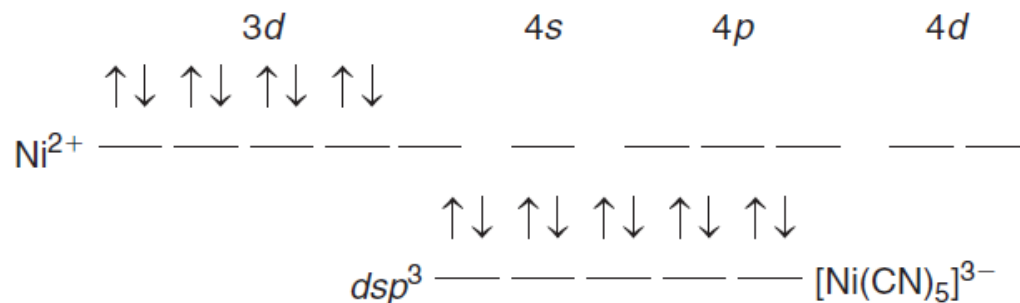


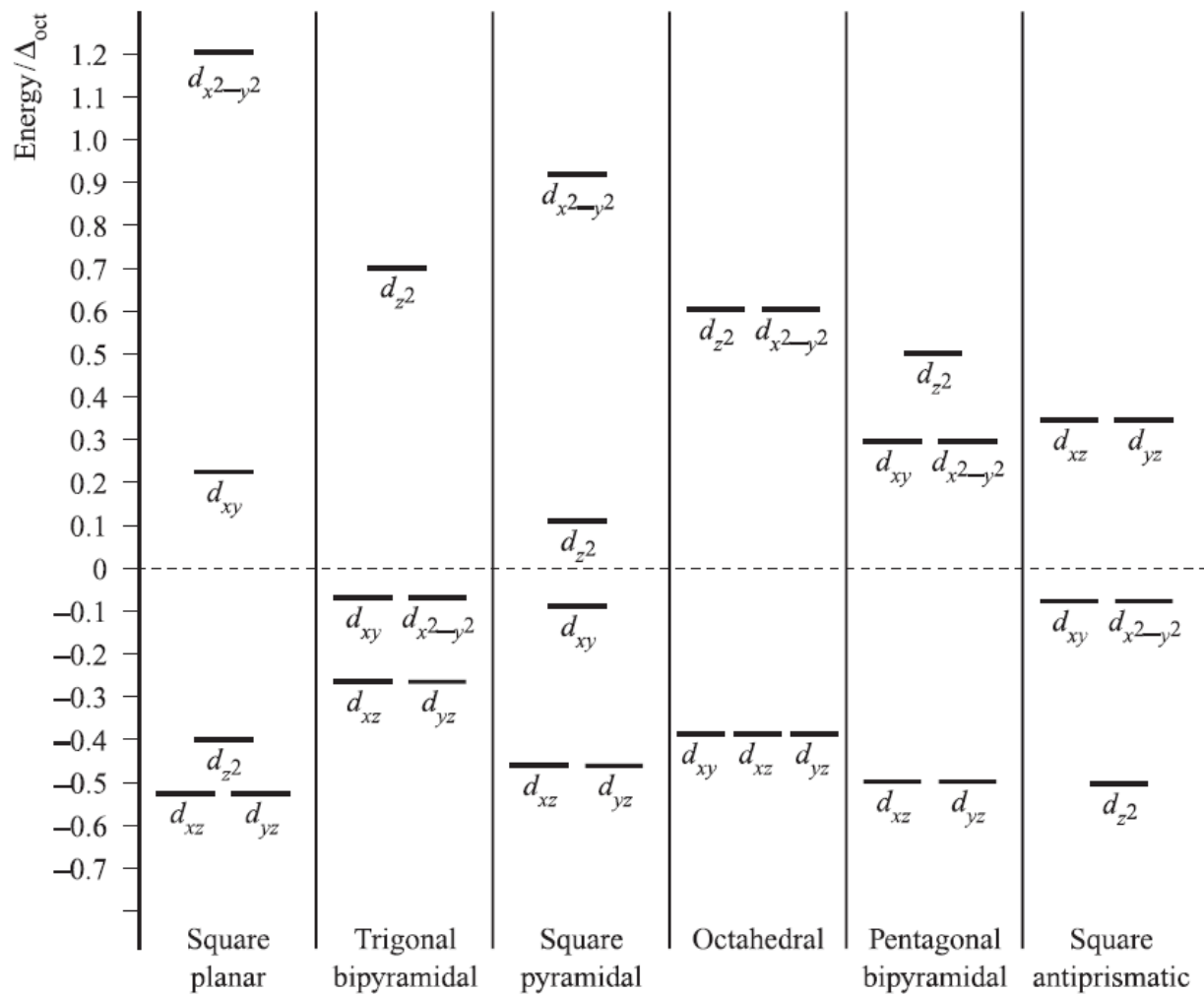
Hybridization

When the d^7 and d^8 electron configurations are considered, there are additional possibilities for forming bonding orbitals. With seven or eight electrons in the five orbitals, there is no possibility of having two of the $3d$ orbitals vacant so if an octahedral complex is formed, it must be of the high-spin, outer-orbital type. Both Co^{2+} (d^7) and Ni^{2+} (d^8) form many complexes of this type. With all of the $3d$ orbitals being populated (not necessarily *filled*), the possibility exists for using the $4s$ and $4p$ orbitals in forming tetrahedral complexes and $[\text{CoCl}_4]^{2-}$ and $[\text{Ni}(\text{NH}_3)_4]^{2+}$ have tetrahedral structures. Although it is not possible to have two of the $3d$ orbitals vacant, it is certainly possible to have one empty, so the possibility of the dsp^3 hybrid orbital type exists. With regard to the orbitals used, the Co^{2+} ion behaves as follows:



For the d^8 Ni^{2+} ion, the orbitals and bonding can be shown as follows:





Crystal field splitting diagrams for some common fields referred to a common barycentre; splittings are given with respect to Δ_{oct} .

Crystal Field Theory (1930)

